

Connecting EXHALE to the Lower Atmosphere: Survey, Implementation, and Example Applications

1. Motivation and current state

EXHALE's lower boundary is the $\sim 1 \mu\text{bar}$ level: the input "Planet radius" is the *radius of that level* (user-guessed), the base temperature is fixed at $T_0 = T_{\text{eq}}$, the base density n_0 is an input (or pressure-anchored via `base_p_ubar`), and the gas is purely atomic H/He + trace metals at prescribed abundances. No molecules exist anywhere in the domain. The Wind-AE IC port carries a *simplified* molecular layer (μ -blend `molec_adjust` + bolometric heating/cooling, auto-shutoff when the atomic transition reaches the base).

This memo (i) surveys how the literature connects escape models to the lower atmosphere, (ii) documents the machinery implemented in `src/modules/lower_atmosphere/` and `src/utis/run_lower.py`, (iii) applies it to the four planets modelled with EXHALE (HD 209458 b, HD 189733 b, WASP-121 b, WASP-52 b), and (iv) prescribes how the outputs are used and connected to EXHALE runs.

2. What the literature does at the interface

Approach	Lower atmosphere	Hands to the wind model at $\sim 1 \mu\text{bar}$	Cost / availability
Salz et al. (2016), TPCI	none — CLOUDY solves the whole column (molecules disabled)	n/a (single domain; base at $n = 10^{14} \text{ cm}^{-3}$, T_{eq})	$\sim 3 \times 10^5$ CPU-h for 18 planets; interface not public
Koskinen et al. (2022)	<i>analytic</i> : isothermal- T_{eq} hypsometric column from 1 bar, $\mu(p)$ from chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ fits (Visscher et al. 2006)	base <i>radius</i> $r_0(1 \mu\text{bar})$, $T_0=T_{\text{eq}}$, $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$; base v from mass conservation	trivial (closed-form; all equations in the paper)
Huang et al. (2017)	<i>analytic spacer</i> : isothermal $\mu=2.3$ column, 1 bar $\rightarrow 10 \mu\text{bar}$	base radius of the atomic layer; $\text{Ly}\alpha$ absorbing bottom boundary (H_2 accidental resonances, $\tau=1$ at $N_{\text{H}_2} \approx 10^{14} \text{ cm}^{-2}$)	trivial
Taylor et al. (2025, 2026)	full Lavvas & Arfaux (2021) model (RC temperature + ~ 150 -species kinetics of Lavvas et al. 2014 + haze microphysics; $10^3 \rightarrow 10^{-6}$ bar)	$T(1 \mu\text{bar})$, altitude of the $1 \mu\text{bar}$ level, composition (with a <i>tolerated discontinuity</i>), K_{zz}	in-house, not public ; coupling is a manual splice, iterated $\sim 3\times$

Key quantitative facts:

- **Base insensitivity for strong winds** (Salz et al. 2016, §3.5–3.6): $T_{\text{base}} \pm 40\% \Rightarrow \dot{M} \pm 5\text{--}10\%$ (linear); base density $10^{14} \rightarrow 10^{12} \text{ cm}^{-3} \Rightarrow \dot{M} \lesssim 40\%$ lower. Their error budget: base density $\pm 50\%$, base $T \pm 10\%$ — both dwarfed by the XUV uncertainty ($\times 3$). $\Rightarrow$ EXHALE’s fixed- T_{eq} base is already defensible for hot Jupiters.
- **Where the fixed base breaks:** (i) planets near radiative equilibrium (stable thermospheres); (ii) cool/sub-Neptune planets where H_2 survives into the wind — Salz’s GJ 1214 b test: H_2 10–25% throughout, H^- a significant coolant, $\dot{M} - 15\%$; Taylor et al. (2026): adding H_2 halves the He 10830 signal (14% $\rightarrow$ 7%); Koskinen et al. (2022) hot Uranus: base $q_{\text{H}_2} \approx 0.84$ and H_3^+ IR emission is the dominant radiative coolant below $\sim 2.3 R_p$.
- **What a photochemical lower model uniquely provides** (Lavvas et al. 2014; Lavvas & Arfaux 2021): the $\text{H}_2 \rightarrow \text{H}$ transition can sit exactly at 1 μbar and is temperature-sensitive there; photochemistry (OH-catalysed H_2 destruction) dissociates H_2 even at 500–1000 K; *atomic-metal release profiles* (Na, K, Mg, Ca, Fe, Al, Si return to atomic form below $\sim 10^{-3}$ bar) — the physically-motivated base abundances for EXHALE’s trace metals; photoionization-dominated electron density ($\sim 10^8 \text{ cm}^{-3}$, $100\times$ Saha); radical/haze heating of $\pm 100\text{--}400$ K right at the handoff level.
- **The published coupling is loose, not monolithic:** Lavvas et al. (2014) take $T(p < 1 \mu\text{bar})$ from the Koskinen thermosphere and use “species < 3 amu escape at the wind velocity” as their upper BC; Taylor et al. tolerate a composition discontinuity at 1 μbar . Only TPCI runs a tight per-timestep coupling (and it disables molecules).

3. The three-tier plan

Tier 1 — analytic lower column (Koskinen-2022 style): derive the 1- μbar base radius and base $\text{H}_2/\text{H}/\text{He}$ from a closed-form column. **Tier 2** — molecular extension of EXHALE (H_2 , H_2^+ , H_3^+ , HeH^+ chemistry + H_3^+ IR cooling), required only for warm Neptunes / sub-Neptunes; **Tier 2a** — an interim EOS-only molecular-base correction. **Tier 3** — consume a real lower-atmosphere model (photochemistry + radiative–convective $T(p)$) through a file interface; the open-source VULCAN (Tsai et al. 2017, 2021) + HELIOS (Malik et al. 2017) stack is the practical replacement for the non-public Lavvas & Arfaux code, and a TPCI-style full coupling is rejected (cost, availability, and demonstrated base insensitivity). Which science case needs which tier: hot Jupiters $\rightarrow$ Tier 1 only; He 10830/ $\text{H}\alpha$ work $\rightarrow$ Tier 1; metal transmission lines $\rightarrow$ Tier 3 (atomic-metal release abundances); warm Neptunes / sub-Neptunes $\rightarrow$ Tier 2 + 1.

4. Implementation

All Fortran lives in `src/modules/lower_atmosphere/`; the Python driver in `src/utils/`. Everything is **opt-in / default off** — with no keys and no `base.inp`, standard runs are unchanged (regression: HD 209458 b He 2^3S +metals 3000-step $\log \dot{M} = 8.97$, identical to EXHALE_v1.0).

4.1 Tier 1: lower_column.f90

Equations. The isothermal hypsometric relation is integrated from the 1-bar level (radius $R_{1\text{bar}}$, in practice the broadband transit radius) to the base pressure $p_0 = 1 \mu\text{bar}$,

$$\frac{d \ln p}{dr} = -\frac{\mu(p, T) m_{\text{H}} g(r)}{k_{\text{B}} T}, \quad g(r) = \frac{GM_{\text{p}}}{r^2}, \quad T = T_{\text{eq}}, \quad (1)$$

with a fixed-step 4th-order Runge–Kutta in $x = \ln p$ (2000 steps; the integrand is smooth). The mean molecular weight follows the chemical-equilibrium H_2/H partition of Visscher et al. (2006) as fitted by Koskinen et al. (2022, Eq. 11),

$$q_{\text{H}_2} \simeq \frac{1.9845 + 10^u - \sqrt{10^u (3.9690 + 10^u)}}{2.3670}, \quad u = -\frac{23672}{T} - \log_{10} p_{[\text{bar}]} + 6.2645. \quad (2)$$

An implementation subtlety uncovered during validation: Eq. (2) returns the *full-mixture* volume mixing ratio $q_{\text{H}_2} = n_{\text{H}_2}/(n_{\text{H}_2} + n_{\text{H}} + n_{\text{He}})$ (a protosolar He abundance is baked into the fit), so the fraction x_2 of H *nuclei* bound in H_2 must be recovered as

$$x_2 = \frac{2q_{\text{H}_2}(1 + f_{\text{He}})}{1 + q_{\text{H}_2}} \leq 1, \quad \mu = \frac{1 + 4f_{\text{He}}}{(1 - x_2) + x_2/2 + f_{\text{He}}} \text{ [amu]}, \quad (3)$$

with $f_{\text{He}} = n_{\text{He}}/n_{\text{H(nuclei)}}$ the elemental ratio (EXHALE’s HeH). Equation (3) was verified against the Koskinen et al. (2022) Model A base composition ($q_{\text{H}_2} = 0.84$, $q_{\text{H}} = 0.026 \Rightarrow x_2 = 0.985$); the naive H-subsystem inversion $x_2 = 2q/(1 + q)$ is *wrong* and misses the gate.

Interface. `lower_column_solve(Mp, R_deep, T, fhe, p_deep, p_base, r_base, qh2_b, qh_b, qhe_b, mu_b, r_base_atomic)` (cgs; pressures in bar). Because chemical equilibrium *underestimates* H dissociation at $T_{\text{eq}} \approx 1000\text{--}2000 \text{ K}$ (photochemistry ignored; Moses et al. 2011; Koskinen et al. 2013a), the routine also integrates the same column with the fully atomic $\mu = (1 + 4f_{\text{He}})/(1 + f_{\text{He}})$ and returns `r_base_atomic`: the two radii *bracket* the truth. The public helper `q_h2_equilibrium(p, T)` exposes Eq. (2) (NaN-safe, clamped to $[0, 1]$).

EXHALE key. Lower column: `<R_1bar in R_J>` in `input.inp` triggers a startup report: the derived $r_0(1 \mu\text{bar})$ (equilibrium + atomic bracket), the input “Planet radius” for comparison, and the base $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}/\mu$, plus a warning when the equilibrium base is strongly molecular. *Report only — nothing is overridden* (overriding is Tier 3’s job, §4.4).

Validation gate (passed). Koskinen et al. (2022) Model A (Uranus-mass planet, $M = 8.6813 \times 10^{28} \text{ g}$, $R_{1\text{bar}} = 2.5559 \times 10^9 \text{ cm}$, Sun-like star at 0.05 au, $T_{\text{eq}} = 1140 \text{ K}$, $f_{\text{He}} = 0.0793$):

	$r_0/R_{1\text{bar}}$	$q_{\text{H}_2}(\text{base})$	$q_{\text{H}}(\text{base})$
this module	1.343	0.838	0.027
Koskinen et al. (2022)	1.34	≈ 0.84	≈ 0.026

Approximations (documented in the source): isothermal column at T_{eq} ; spherical gravity (the Roche-equipotential correction matters only near RLO); chemical equilibrium only.

4.2 Tier 2a: the Molecular base key

Molecular base: True applies an *EOS-only* correction at startup: the H nuclei bound into H₂ at (1 μbar, T₀) per Eq. (2)–(3) are removed from the base particle budget,

$$\text{ntot_bc} \rightarrow \text{ntot_bc} - \frac{x_2/2}{1 + f_{\text{He}}}, \quad (4)$$

which lowers the base pressure ($p_{\text{base}} = \text{ntot_bc} n_0 k_B T_0$) and raises the base mean molecular weight, i.e. a heavier, more compact base — the leading-order effect of a molecular base on the hydrodynamics. The species arrays remain atomic (H₂ chemistry is Tier-2 proper), so this is a *crude, clearly-flagged* approximation. Example: HD 209458 b, $q_{\text{H}_2}(1 \mu\text{bar}, 1450 \text{ K}) = 0.831 \Rightarrow \text{ntot_bc} 1.0 \rightarrow 0.546$.

4.3 Tier 2 foundation: h3p_cooling.f90 and mol_rates.f90

H₃⁺ infrared cooling. Optically thin cooling per molecule from Miller et al. (2013, Table 5, $z(T)$ fits, archived as references/Miller_2013_JPCA_117_9770.pdf):

$$\ln E(\text{H}_3^+, T) = \sum_n C_n T^n \quad [E \text{ in W molecule}^{-1} \text{ sr}^{-1}], \quad (5)$$

piecewise over 30–300, 300–800, 800–1800, and 1800–5000 K (fit errors < 0.1% over 300–5000 K), held frozen above 5000 K (H₃⁺ is destroyed well below). The volumetric rate is

$$\Lambda_{\text{H}_3^+} = n_{\text{H}_3^+} 4\pi E(T) s_{\text{nonLTE}}(T, n_{\text{H}_2}) \times 10^7 \quad [\text{erg s}^{-1} \text{ cm}^{-3}], \quad (6)$$

where s_{nonLTE} is the departure factor of Miller et al. (2013, Table 6; their Oka & Epp detailed-balance approach), implemented as a bilinear interpolation in $(T, \log_{10} n_{\text{H}_2})$ over $T = 300\text{--}5000 \text{ K}$ and $n_{\text{H}_2} = 10^6\text{--}10^{14} \text{ cm}^{-3}$; $s \rightarrow 1$ (LTE) at high density, and the tabulated values are upper limits per the paper’s caveat on the proton-hopping rate. Both interpolators use the NaN-safe `.not.(x>edge)` bracketing adopted after the 2026-07-02 code review. **Validation:** the fits reproduce the paper’s Table-4 anchors at $T = 500/1000/2000/3000/5000 \text{ K}$ to <0.5%, and $s(1000 \text{ K}, 10^{10} \text{ cm}^{-3}) = 0.4955$, $s(2000 \text{ K}, 10^{12}) = 0.9499$ exactly.

Molecular reaction rates. `mol_rates.f90` provides the full H₂/H₂⁺/H₃⁺/HeH⁺ thermochemical network of Koskinen et al. (2022, Table 1; transcribed and verified against the PDF page): recombination (R1–R2, Storey & Hummer 1995), Voronov (1997) collisional ionization (R3–R4), H₂⁺/H₃⁺/HeH⁺ dissociative recombination (R5–R7, R16; Larsson et al. 2008), the H₃⁺ production backbone H₂⁺ + H₂ → H₃⁺ + H (R8, 2×10^{-9}), thermal and electron-impact H₂ dissociation (R12, R14), three-body chemistry (R13, R15), the HeH⁺ chain (R17–R20), and H/He charge exchange (R21–R23). The stellar photo-rates P1–P5 (including H₂ dissociative and double photoionization) follow EXHALE’s existing PH_heat machinery once H₂ cross sections are added. Baseline caveat inherited from the source: Lyman–Werner neutral photodissociation is not in the network (Koskinen et al.’s sensitivity test bounded its effect on $\dot{M}$ at $\times 1.4$).

Tier-2 core (implemented 2026-07-03). The coupled system `System_HeH_mol.f90` solves 7–8 unknowns per cell (H⁺, He⁺, He⁺⁺, optionally He 2³S, plus the H-nucleus fractions bound in H₂, H₂⁺, H₃⁺, HeH⁺) as element-conserving

steady-state balances with `hybrd1`: the atomic rows use EXHALE’s own rate arrays (so the molecule-free limit reproduces the atomic systems’ solution) and the molecular channels are the Table-1 network via `mol_rates` (R21/R22 H–He charge exchange deliberately excluded to preserve the atomic limit). The H_2 photoionization cross section (Yan et al. 1998, Eqs. 17–19; verified against their Table 7 to $<0.2\%$) enters the opacity, the P_{H_2} photoionization integral, the photoelectric heating, and the absorbed-energy normalization through optional arguments of `PH_heat_HHe`; H_3^+ IR cooling enters the `cool` array. The EOS counts each molecule as one particle, each molecular ion’s electron, and the correct masses (`calc_ne/ntot/rho`, `get_species_densities`). `f_sp` grows to 37 columns (output schema and Load-IC names extended). Key: Molecular chemistry: True (requires He; v1 excludes trace metals; pair with Molecular base: True for a consistent ghost base pressure). **Gates passed:** (0) molecules-off regression, HD 209458 b 3000-step $\log \dot{M} = 8.97$ unchanged; (1) HD 209458 b molecules-on: a fully molecular base with a sharp $\text{H}_2 \rightarrow \text{H}$ front at $r = 1.019R_p$ and an essentially atomic wind above it ($\log \dot{M}$ 8.91 vs. 8.97; He 2^3S peak +1.4%) — the hot-Jupiter atomic base is now a *result*, not an assumption; (2) a hot-Uranus-like case ($0.0457 M_J$, $T_{\text{eq}}=1140\text{ K}$): the front rises to $r = 1.148R_p$ and H_3^+ is active below it. Figure 1 shows the gate runs.

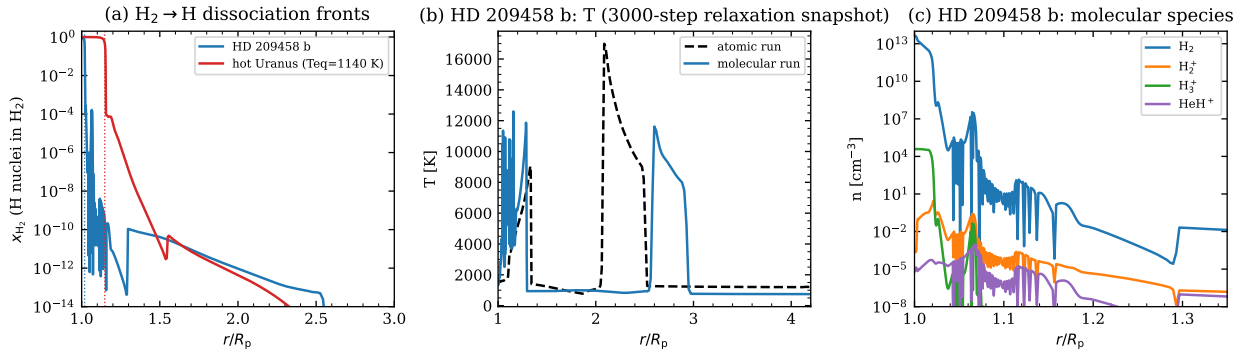


Figure 1: Tier-2 molecular gates (3000-step relaxation snapshots). (a) The H nuclei fraction bound in H_2 : on HD 209458 b the base is fully molecular and a sharp $\text{H}_2 \rightarrow \text{H}$ front sits at $r = 1.019R_p$ (dotted); on the cooler, low-gravity hot Uranus the front rises to $1.148R_p$. The cell-to-cell flip-flop just above the front marks the chemically bistable zone of *equilibrium* H_2 chemistry — kinetics+advection smooth it (the documented Tier-2 caveat). (b) The molecular run’s temperature agrees with the atomic run away from the (still-relaxing) front region. (c) Molecular species on HD 209458 b: H_3^+ (green) and the molecular ions are confined below the front, as expected for a hot Jupiter.

Remaining Tier-2 work (open). Molecular *advection* (the local chemical-equilibrium treatment cannot reproduce Koskinen et al.’s advective replenishment of high-altitude H_2 on cool Neptunes); Lyman–Werner photodissociation ($\lesssim \times 1.4$ on $\dot{M}$ per their test); the dissociative/double photoionization channels (P4/P5); the 4.48 eV dissociation energy sink; a diatomic γ ; the metals+molecules merge; and the GJ 1214 b He-halving validation (Taylor et al. 2026) with a full M-dwarf setup.

4.4 Tier 3: base.inp and run_lower.py

The handoff file. An optional base.inp in the run directory is read by input_read *before* the derived constants, so its overrides propagate consistently (v_0 , b_0 , grid, normalization):

key	unit	overrides	used by
T_base	K	T_0 (base temperature)	thermal BC, EOS scales
r_base	R_J	“Planet radius” ($= r_0$ of 1 μ bar)	grid, gravity, areas
HeH_base	–	He/H number ratio	composition, He diffusion
Kzz_base	cm^2/s	he_kzz	He_diffusion homopause

Every override is echoed to stdout; an absent file is a strict no-op. Unknown keys and # comments are ignored, so a richer lower-atmosphere model may carry extra metadata in the same file.

The driver. `src/utis/run_lower.py <run_dir> --r1bar <R_J> [--guillot] [--tint T] [--kzz K]` reads M_p , T_{eq} , He/H from the run directory’s input.inp, integrates the same column as §4.1 in Python, and writes base.inp. Two temperature models: isothermal T_{eq} (default; identical to the Fortran Tier-1 column) and the Guillot (2010) semi-grey global-average profile

$$T^4(\tau) = \frac{3T_{\text{int}}^4}{4} \left(\frac{2}{3} + \tau \right) + \frac{3T_{\text{eq}}^4}{4} \left(\frac{2}{3} + \frac{1}{\gamma\sqrt{3}} + \left[\frac{\gamma}{\sqrt{3}} - \frac{1}{\gamma\sqrt{3}} \right] e^{-\gamma\tau\sqrt{3}} \right), \quad \tau = \kappa_{\text{IR}} p/g, \quad (7)$$

with standard hot-Jupiter defaults ($\kappa_{\text{IR}} = 10^{-2} \text{ cm}^2 \text{ g}^{-1}$, $\gamma = 0.4$, $T_{\text{int}} = 100 \text{ K}$). The file records the base $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$ as comments and prints a molecular-base warning when $q_{\text{H}_2} > 0.3$. The upgrade path — running VULCAN for the composition and a real RC model for $T(p)$ — writes the *same* base.inp keys, so nothing on the EXHALE side changes.

VULCAN upgrade implemented and demonstrated (2026-07-03). The public VULCAN code (cloned to `./VULCAN`; FastChem compiled) was run to steady state for HD 189733 b (SNCHO-2025 photochemical network, 2356 steps) and converted with the new `src/utis/vulcan_to_base.py` (`.vul` $\rightarrow$ `base.inp`: photochemical $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$ at 1 μ bar, hypsometric r_{base} with VULCAN’s own $\mu(p)$ and $T(p)$, molecular mixing ratios as comments). **Result:** at 1 μ bar the photochemical base is $q_{\text{H}_2} = 0.63$, $q_{\text{H}} = 0.23$ — *photochemistry dissociates $\sim 11\times$ more H than the chemical-equilibrium column* ($q_{\text{H}} = 0.020$), directly quantifying the Tier-1 caveat (Moses 2011; Koskinen 2013a); $r_{\text{base}} = 1.168$ vs the analytic 1.174 R_J ; $T_{\text{base}} = 863 \text{ K}$ (the Moses11 $T(p)$, cooler than the isothermal $T_{\text{eq}} = 1183 \text{ K}$). The full chain `VULCAN` $\rightarrow$ `converter` $\rightarrow$ `base.inp` $\rightarrow$ EXHALE startup override was exercised end-to-end. Caveat: VULCAN’s networks are H/C/N/O(S) — the atomic-metal release (Na/Mg/Ca/Fe) has no public analogue and remains with the Lavvas collaboration path; `metals.inp` stays user-supplied. Figure 2 shows the converged composition against the equilibrium H_2 track.

5. Example applications: HD and WASP planets

The machinery was applied to the four planets modelled with EXHALE (`examples/13_lower_atmosphere/`; $R_{1\text{bar}} =$ broadband transit radius; T_{eq} , M_p , He/H from each run’s input.inp).

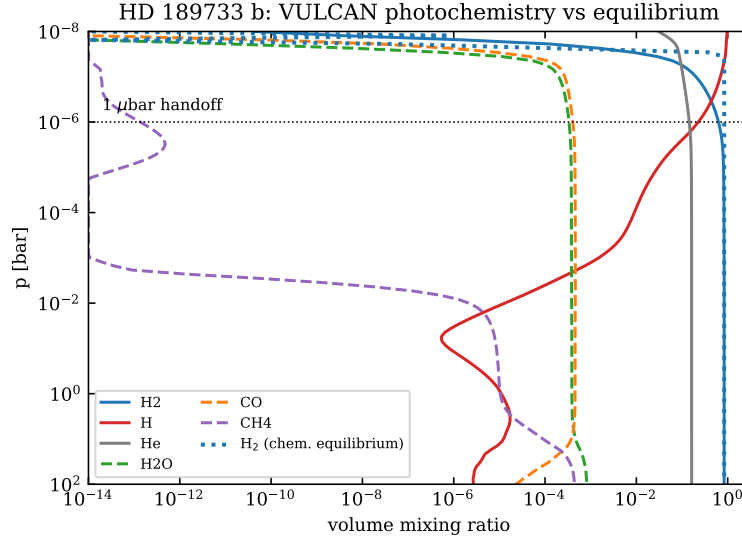


Figure 2: HD 189733 b lower/middle atmosphere from the converged VULCAN run (solid/dashed: H_2 , H , He , H_2O , CO , CH_4) compared with the chemical-equilibrium H_2 mixing ratio evaluated along the same $T(p)$ (dotted). Above $\sim 10^{-4}$ bar photochemistry dissociates H_2 far beyond equilibrium; at the $1\ \mu\text{bar}$ handoff (horizontal dotted line) $q_{\text{H}} = 0.23$ versus the equilibrium 0.020 — the $\sim 11\times$ underestimate quantified.

	M_{p}	T_{eq}	He/H	$R_{1\text{bar}}$	input R_0	$r_0(1\ \mu\text{bar}) [R_{\text{J}}]$	r_0^{Gui}	q_{H_2}	μ_{base}
planet	$[M_{\text{J}}]$	$[\text{K}]$		$[R_{\text{J}}]$	$[R_{\text{J}}]$	chem.eq. atomic	$[R_{\text{J}}]$	(base)	[amu]
HD 209458 b	0.720	1450	0.0833	1.36	1.401	1.472 1.582	1.466	0.831	2.25
HD 189733 b	1.237	1183	0.0833	1.138	1.193	1.174 1.205	1.171	0.838	2.26
WASP-121 b	1.182	2358	0.0851	1.753	2.2075	2.010 2.133	1.966	0.005	1.24
WASP-52 b	0.460	1304	0.0204	1.27	1.270	1.437 1.615	1.428	0.837	1.95

Isothermal- T_{eq} column (chem. equilibrium + fully-atomic bracket) and the Guillot variant r_0^{Gui} ($T_{\text{base}}^{\text{Gui}} = 1313, 1072, 2136, 1181\ \text{K}$ respectively). “input R_0 ” is the base radius currently used in the production runs.

Interpretation (per planet):

- **HD 189733 b — consistent.** The production base radius ($1.193 R_{\text{J}}$) falls *inside* the derived bracket $[1.174, 1.205]$: the adopted setup is validated by the column, not merely assumed.
- **HD 209458 b — input $\sim 5\%$ low.** The production value $1.401 R_{\text{J}}$ sits just below the bracket $[1.472, 1.582]$. Given the isothermal- T_{eq} assumption this is within the method’s spread, but it suggests the $1\text{-}\mu\text{bar}$ level may lie slightly higher than assumed; the effect on $\dot{M}$ is bounded by the Salz base-sensitivity results ($\lesssim 10\%$ -level).
- **WASP-121 b — the atomic base is *verified*.** At $T_{\text{eq}} = 2358\ \text{K}$ the equilibrium base is already atomic ($q_{\text{H}_2} = 0.005$, $q_{\text{H}} = 0.92$, $\mu = 1.24$): for this ultra-hot Jupiter EXHALE’s atomic assumption holds *even in equilibrium*, with no appeal to photochemistry. The production base radius $2.2075 R_{\text{J}}$ (derived earlier from the Huang et al. 2023 Table-3 $\log g$ anchoring) lies $\sim 4\text{--}10\%$ above the isothermal- T_{eq} bracket $[2.010, 2.133]$ — consistent with a dayside lower atmosphere hotter than the redistribution-averaged T_{eq} , which extends the column. Note the deep

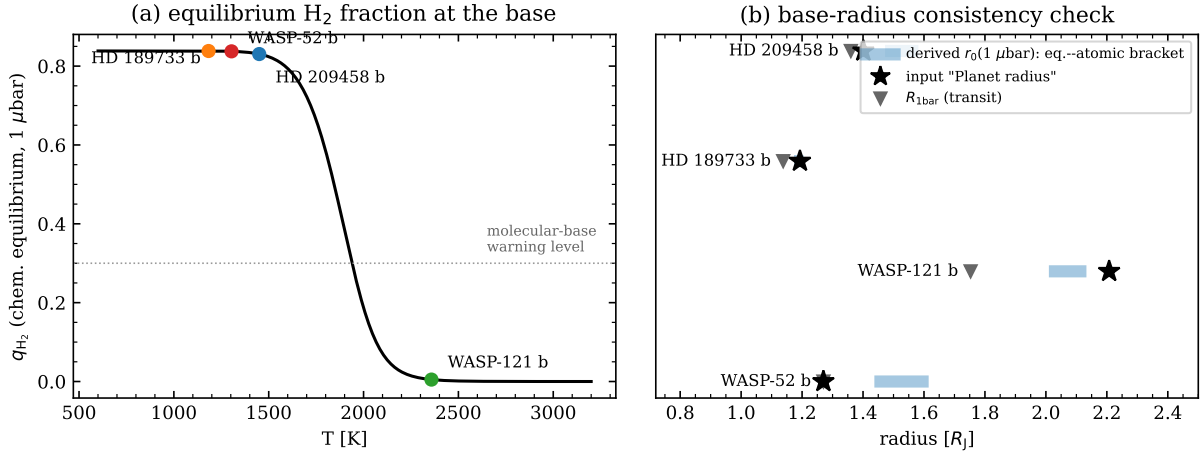


Figure 3: Tier-1 analytic column applied to the four planets. (a) The chemical-equilibrium H_2 mixing ratio at 1 μ bar (Eq. 2) as a function of temperature: the three $T_{eq} \lesssim 1500$ K planets sit on the fully molecular plateau, while WASP-121 b ($T_{eq} = 2358$ K) is atomic even in equilibrium. (b) Derived 1- μ bar base radius (equilibrium-atomic bracket, shaded) versus the production “Planet radius” (star) and the transit radius (triangle): HD 189733 b falls inside its bracket, HD 209458 b sits $\sim 5\%$ below, WASP-121 b lies just above (dayside-hot lower atmosphere), and WASP-52 b’s transit-radius shortcut puts the base 0.17–0.35 R_J too deep.

layers are still molecular ($q_{H_2}(1 \text{ bar}) \approx 0.83$), so μ varies from 2.25 to 1.24 along this column — the equilibrium track is *shorter* than the atomic bracket here because of its heavier deep-layer μ .

- **WASP-52 b — misplaced base, impact quantified.** The production runs use the transit radius itself ($1.27 R_J$) as the 1- μ bar base — the common shortcut — but the column places that level at $1.44\text{--}1.62 R_J$, i.e. the base is $0.17\text{--}0.35 R_J$ (13–27%) too deep. *Recomputation* (the fxuv1p0_he98 configuration rerun Newton-converged with the corrected equilibrium-column base $r_0 = 1.437 R_J$, all else identical): $\log \dot{M} \ 11.97 \rightarrow 12.14$ ($\times 1.5$), but the He 10830 observable barely moves — line-center absorption $27.11 \rightarrow 27.56\%$ ($\times 1.02$) and equivalent width $0.713 \rightarrow 0.722 \text{ \AA}$ ($\times 1.01$). The line-forming effective radius is set by the $\tau=1$ surface of the extended wind, which stays nearly fixed in *physical* units (the larger, less-bound base raises $\dot{M}$, but the He 2^3S column at a given physical radius is almost unchanged: peak $n_{2^3S} \ 150.5 \rightarrow 147.6 \text{ cm}^{-3}$). Conclusion: the base misplacement biases the *mass-loss rate* at the $\times 1.5$ level but appears not to bias the He 10830 scan conclusions materially. Also note this run family’s low He/H ($= 0.0204$, the “He98” abundance scan), visible in the light μ_{base} .
- **Common pattern.** All three $T_{eq} \lesssim 1500$ K planets have strongly molecular *equilibrium* bases ($q_{H_2} \approx 0.83$, the fully-molecular limit of the fit) — quantifying that their atomic bases rest on *photochemical* dissociation (Moses et al. 2011; Koskinen et al. 2013a) rather than thermal, which is precisely what the Tier-3 VULCAN upgrade would supply from first principles. The Guillot column is systematically slightly more compact (cooler skin above the photosphere) — a $\lesssim 0.5\%$ radius effect at fixed assumptions.

6. Using and connecting the results

6.1 Recommended per-planet workflow

1. **Column check (always, free).** Add `Lower` column: `<transit radius>` to `input.inp` (or run `run_lower.py`).
Read off: does the current “Planet radius” fall inside the `[chem.eq., atomic]` bracket? Is $q_{\text{H}_2}(\text{base})$ small?
2. **Decide the tier.** $q_{\text{H}_2} \lesssim 0.1$ (e.g. WASP-121 b): the atomic model is verified — proceed as usual. $q_{\text{H}_2} \sim 0.8$ with $T_{\text{eq}} \sim 1000\text{--}2000$ K (typical hot Jupiters): the atomic base rests on photochemistry — proceed, but note the assumption; a genuinely cool planet (warm Neptune / sub-Neptune) needs Tier-2 physics before its He 10830 predictions are trusted.
3. **Adopt the geometry.** If the input radius lies outside the bracket (WASP-52 b), either correct “Planet radius” by hand or copy the driver’s `base.inp` into the run directory so EXHALE adopts r_0 , T_0 , He/H and K_{zz} automatically (echoed at startup).
4. **Run EXHALE** with the usual two-stage convergence (+ Newton finish for quantitative $\dot{M}$), then `TPM.py` for the transit spectra.
5. **(Optional) iterate** once a real lower-atmosphere model is attached (§6.3).

6.2 Where each quantity connects

quantity	produced by	consumed by / effect
$r_0(1\ \mu\text{bar})$	Tier 1 / <code>base.inp</code>	grid inner radius, gravity b_0 , irradiated area; TPM continuum radius
bracket $[r_0^{\text{eq}}, r_0^{\text{atom}}]$	Tier 1	consistency check on the input radius; systematic-error bar on r_0
T_{base}	driver (T_{eq} or Guillot)	thermal BC T_0 ; v_0 , p_0 , b_0 normalization
$q_{\text{H}_2}(\text{base})$	Eq. (2)	tier-decision switch; Molecular base correction; Tier-2 initial condition
He/H (base)	<code>base.inp</code>	composition; the He_diffusion reservoir value
K_{zz}	<code>base.inp</code>	he_kzz: homopause of the He/H (and metal) diffusive separation
$\Lambda_{\text{H}_3^+}(T, n_{\text{H}_3^+}, n_{\text{H}_2})$	<code>h3p_cooling</code>	(Tier 2) new term in the cool array of the energy equation
R1–R23 rates	<code>mol_rates</code>	(Tier 2) source terms of <code>System_HeH_mol</code> ; $\text{H}_3^+/\text{HeH}^+$ densities for the cooling and for IR-diagnostic post-processing
metal release n_X/n_{H}	(Tier 3, VULCAN)	<code>metals.inp</code> values — replaces assumed solar totals for the Mg II/Ca II/Na I transmission lines

6.3 Feedback loop (future, with a real lower model)

One-way coupling (`lower` → `base.inp` → EXHALE, with the Taylor-style tolerated composition discontinuity) is the default. For the two-way iteration: EXHALE already computes the attenuated stellar spectrum on its energy grid, so the *transmitted XUV flux at the base*, $F_\nu(r_0) = F_\nu e^{-\tau_\nu(r_0)}$, can be dumped and handed to the photochemical model as its top irradiation; and EXHALE’s $\dot{M}$ supplies the lower model’s upper BC (“species < 3 amu escape at the wind velocity”, exactly the Lavvas et al. 2014 prescription). Taylor et al. found ~ 3 iterations sufficient for the analogous $\text{Ly}\alpha$ loop; the same is expected here since the wind is insensitive to base details at the $\lesssim 10\%$ level (Salz et al. 2016).

6.4 Consistency cautions

- Do not double-count composition: if `base.inp` sets He/H from a lower model, the `metals.inp` abundances should be the same model's *atomic release* values, not solar totals.
- `base.inp` overrides `win` over `input.inp` for T_0 , R_0 , He/H (echoed); remove the file to return to the manual values.
- The Tier-1 column is isothermal and spherical: near-RLO planets (WASP-121 b) need the Roche correction before the bracket is taken quantitatively, and dayside-hot lower atmospheres extend beyond the T_{eq} column (both visible in the WASP-121 b example above).
- The `Molecular base` key changes only the EOS budget; do not combine it with quantitative He 10830 claims for genuinely cool planets — that is Tier-2 physics.

References

- [1] Banks, P. M., & Kockarts, G. 1973, *Aeronomy* (New York: Academic Press)
- [2] Guillot, T. 2010, *A&A*, 520, A27
- [3] Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, *ApJ*, 851, 150
- [4] Huang, C., Koskinen, T., Lavvas, P., & Fossati, L. 2023, *ApJ*, 951, 123
- [5] Koskinen, T. T., Harris, M. J., Yelle, R. V., & Lavvas, P. 2013a, *Icarus*, 226, 1678
- [6] Koskinen, T. T., Lavvas, P., Huang, C., et al. 2022, *ApJ*, 929, 52
- [7] Larsson, M., McCall, B. J., & Orel, A. E. 2008, *Chem. Phys. Lett.*, 462, 145
- [8] Lavvas, P., Koskinen, T., & Yelle, R. V. 2014, *ApJ*, 796, 15
- [9] Lavvas, P., Koskinen, T., Steinrueck, M. E., García Muñoz, A., & Showman, A. P. 2019, *ApJ*, 878, 118
- [10] Lavvas, P., & Arfaux, A. 2021, *MNRAS*, 502, 5643
- [11] Malik, M., Grosheintz, L., Mendonça, J. M., et al. 2017, *AJ*, 153, 56
- [12] Miller, S., Stallard, T., Tennyson, J., & Melin, H. 2013, *J. Phys. Chem. A*, 117, 9770
- [13] Moses, J. I., Visscher, C., Fortney, J. J., et al. 2011, *ApJ*, 737, 15
- [14] Neale, L., Miller, S., & Tennyson, J. 1996, *ApJ*, 464, 516
- [15] Salz, M., Czesla, S., Schneider, P. C., & Schmitt, J. H. M. M. 2016, *A&A*, 586, A75
- [16] Storey, P. J., & Hummer, D. G. 1995, *MNRAS*, 272, 41
- [17] Taylor, A. R., Koskinen, T., Argenti, L., et al. 2025, *ApJ*, 989, 68
- [18] Taylor, A. R., Koskinen, T., Huang, C., Arfaux, A., & Lavvas, P. 2026, *ApJ*, 999, 214

- [19] Tsai, S.-M., Lyons, J. R., Grosheintz, L., et al. 2017, ApJS, 228, 20
- [20] Tsai, S.-M., Malik, M., Kitzmann, D., et al. 2021, ApJ, 923, 264
- [21] Visscher, C., Lodders, K., & Fegley, B. 2006, ApJ, 648, 1181
- [22] Voronov, G. S. 1997, At. Data Nucl. Data Tables, 65, 1
- [23] Xing, L., Yan, D., & Guo, J. 2023, ApJ, 953, 166